Name: ____________________________ Date: ________________ Period: _________

Conic Sections Practice — Key

1. Identify the conic type, center or vertex, and key measurements.

(a) $y + 1 = \frac{1}{2}(x-3)^2$

Type: **parabola** Vertex: $(3, -1)$ $a = \frac{1}{2}$ Opens: **upward**

(b) $\dfrac{(x+2)^2}{36} + \dfrac{(y-4)^2}{16} = 1$

Type: **ellipse** Center: $(-2, 4)$ $a = 6$ $b = 4$

(c) $\dfrac{(x-5)^2}{4} - \dfrac{(y+3)^2}{25} = 1$

Type: **hyperbola** Center: $(5, -3)$ $a = 2$ $b = 5$

2. Write the standard form equation.

(a) Parabola, vertex $(0, 4)$, opening downward, through $(2, 0)$.

$y - 4 = a(x-0)^2$. **Substitute** $(2,0)$**:** $-4 = 4a \Rightarrow a = -1$. **Equation:** $y - 4 = -(x)^2$.

(b) Parabola, vertex $(3, -1)$, opening left, through $(1, 3)$.

$x - 3 = a(y+1)^2$. **Substitute** $(1,3)$**:** $-2 = 16a \Rightarrow a = -\frac{1}{8}$. **Equation:** $x - 3 = -\frac{1}{8}(y+1)^2$.

(c) Ellipse centered at $(-1, 2)$, horizontal radius 6, vertical radius 4.

$$\frac{(x+1)^2}{36} + \frac{(y-2)^2}{16} = 1$$

3. Hyperbola centered at $(4, -3)$, $a = 5$, $b = 2$, opening up and down.

$$\frac{(y+3)^2}{25} - \frac{(x-4)^2}{4} = 1$$

Asymptotes: $y + 3 = \pm\dfrac{5}{2}(x-4)$

4. $9x^2 - 4y^2 = 36$.

(a) **Divide by 36:** $\dfrac{x^2}{4} - \dfrac{y^2}{9} = 1$.

(b) **Center:** $(0, 0)$**;** $a = 2$**,** $b = 3$**; opens left and right. Asymptotes:** $y = \pm\dfrac{3}{2}x$.

AP-Style Practice — Key

5. Correct answer: **(B)**

$$\frac{(x-2)^2}{9} - \frac{(y+1)^2}{16} = 1$$

Teacher Note

(A) is an ellipse (uses $+$, not $-$). (B) is correct: left/right hyperbola has positive x-term; center $(2, -1)$ gives $(x-2)$ and $(y+1)$; $a^2 = 9$, $b^2 = 16$. (C) opens up/down (positive y-term). (D) has wrong center signs: $h = 2$ requires $(x-2)$, not $(x+2)$.